Length of a Module

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The length of a module is a generalization of the dimension of a vector space.

Lemma 0.1. If $l(M) \leq n$, M is generated by at most n elements.

Proof. We induct on the length. If the length is 0 or 1, the result is immediate. Assume the result holds for some $n - 1$ where $n \geq 1$.

Let M be a module such that the length of M is at most n. Then, M has a composition series of length n. Then, M contains a submodule N of length at most $n - 1$ such that M/N is simple.

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